

Retrieval Augmented Generation (RAG)

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What We'll Cover

- Motivating RAG
- Introduction to RAG
- Elaborate retrieval phase
- Common problems of RAG
- RAG variants
- RAG assessment framework



Why RAG?

Limitations with LLMs

LLMs can do a lot of tasks...

Summarize Text

Generate Code

Rewrite Content

...but they suffer from

Hallucinations

Verifiability issues

Knowledge cut-offs

Learning failures

Limitations with LLMs: Hallucinations



What is the use of PMI of NLP?



In natural language processing (NLP), Pointwise Mutual Information (PMI) is often used as a measure of the association between words in a corpus. It can be used to identify words that are highly correlated and occur frequently together, which can be useful...



Can you give me citation for the above examples?



Here are some references for the examples of using PMI in natural language processing (NLP) that I provided:

Language modeling: "Pointwise Mutual Information (PMI) for Language Modeling" by R. B. Frazer and J. H. Lafferty (<https://www.aclweb.org/anthology/W00-1308.pdf>)

Information retrieval: "Information Retrieval using PMI-based Word Representations" by T. Mitamura and K. Kise (<https://www.aclweb.org/anthology/C04-1100.pdf>)

Machine translation: "PMI-based Word Alignment for Statistical Machine Translation" by K. Toutanova and C D. Manning (<https://www.aclweb.org/anthology/W03-0419.pdf>)

Limitations with LLMs: Hallucinations



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Pointwise Mutual Information (PMI) for Language Modeling 

About 13,000 results (0.24 sec)

C-PMI: Conditional pointwise mutual information for turn-level dialogue evaluation
[L Ren](#), [M Sidhu](#), [Q Zeng](#), [RG Reddy](#), [H Ji](#)... - Proceedings of the ..., 2023 - [aclanthology.org](#)
 ... For each of the dialogue turn t , our **Pointwise Mutual Information (PMI)** based metric is con...
language model to obtain the log-likelihood for calculating both the FED score and our C-PMI ...
 ☆ Save  Cite Cited by 10 Related articles All 6 versions 

Unsupervised extractive summarization using pointwise mutual information
[V Padmakumar](#), [H He](#) - Proceedings of the 16th Conference of the ..., 2021 - [aclanthology.org](#)
 ... **pointwise mutual information (PMI)** between sentences, which can be easily computed by a pre-trained **language model**... We use the **language model** to compute **PMI**, which then scores ...
 ☆ Save  Cite Cited by 61 Related articles All 5 versions 

[PDF] PMIVec: a word embedding model guided by point-wise mutual information criterion
[M Yao](#), [L Zhuang](#), [S Wang](#), [H Li](#) - Multimedia Systems, 2022 - [staff.ustc.edu.cn](#)
 ... words, this paper proposes a **point-wise mutual information (PMI)**-guided word embedding ...
 cific, BERT will estimate a masked **language model** during training time, and any **language** ...
 ☆ Save  Cite Cited by 2 Related articles All 3 versions Web of Science: 1 

Limitations with LLMs: Hallucinations



What is the use of PMI of NLP?



In natural language processing (NLP), Pointwise Mutual Information (PMI) is used as a measure of the association between words. It is particularly useful for identifying words that are highly correlated and occur frequently together in a corpus.



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Information Retrieval using PMI-based Word Representations 

About 1,280 results (0.18 sec)

[A Comparative Study of PMI and its Variants for Meaningful Word Selection](#)
[A Toprak, FS Toprak - 2025 Innovations in Intelligent Systems ..., 2025 - ieeexplore.ieee.org](#)
 ... PMI implicitly to enhance word representations. Levy and ... PMI-based methods have been applied in Turkish NLP ... set of annotations using standard information retrieval metrics. ...
 ☆ Save  Cite Related articles All 2 versions

[A comparison of correspondence analysis with PMI-based word embedding methods](#)
[Q Qi, A Bagheri, DJ Hessen... - arXiv preprint arXiv ..., 2024 - arxiv.org](#)
 ... such word representations are related to the factorization of the pointwise mutual information (... This paper focuses on traditional PMI-based word embedding methods. To enhance the ...
 ☆ Save  Cite Cited by 1 Related articles All 6 versions 

[\[PDF\] A comprehensive analysis of PMI-based models for measuring semantic differences](#)
[T Aida, M Komachi, T Ogiso, H Takamura... - ..., language, information ..., 2021 - aclanthology.org](#)
 ... with semantic differences across corpora is mainly addressed by word representations such as word2vec or BERT. However, in the real world where linguists and sociologists apply ...
 ☆ Save  Cite Cited by 31 Related articles All 3 versions 

Limitations with LLMs: Hallucinations



Dangers of hallucination:

- Recipe with incompatible ingredients
- Medical misguidance
- Misidentification of military targets
- Fabricated engineering designs with invalid parameters
- Altering or deleting production databases with confidence

Limitations with LLMs: Verifiability



What would have Richard Nixon said if the Apollo landing failed?



There's no historical record of a "backup" speech for a failed Apollo 11 landing, so anything here is necessarily speculative—but we can infer what a statement from Richard Nixon would likely have emphasized based on how NASA and the White House prepared for worst-case scenarios.

...

A plausible version of his remarks might have been:

"Today, we have suffered a tragedy in our quest to reach the Moon. The astronauts of Apollo 11 set out in the noblest tradition of exploration, and they did so on behalf of all mankind.

Their courage and sacrifice will not be forgotten. In them, we see the very best of human determination and spirit.

Though this mission has ended in sorrow, the United States will continue to pursue space exploration, not in recklessness, but in honor of those who dared greatly."

...

+ Ask anything



Voice

Disclaimer by OpenAI

ChatGPT is AI and can make mistakes.

Limitations with LLMs: Verifiability



Dangers with verifiability:

- Distorted historical narratives (historians believe that...)
- Fabricated expertise (according to my analysis...)
- Personal manipulation (based on your habits...)
- Religious or spiritual claims
- Political manipulation

Limitations with LLMs: Knowledge-cutoff



Who is the winner of IPL 2026?



I don't know who won the IPL 2026 title.

My knowledge doesn't include verified real-world results from events that may have occurred after my training data, and I don't have live access to sports results in this conversation.

If the IPL 2026 season has already concluded, you can check the official IPL website or a sports news source, and I can help analyze the tournament, teams, or statistics if you share the result.

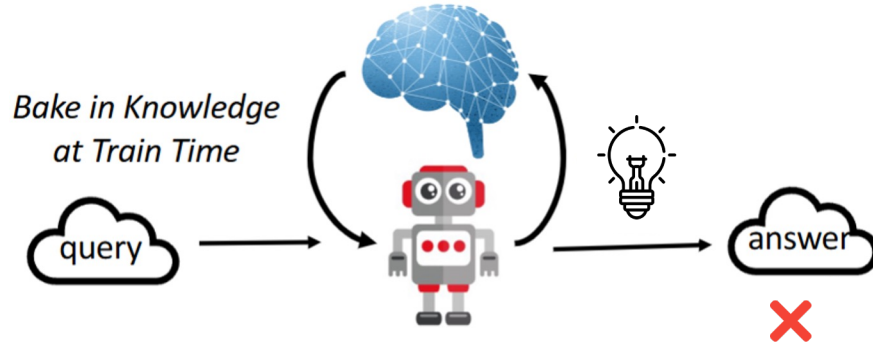
Limitations with LLMs: Knowledge-cutoff



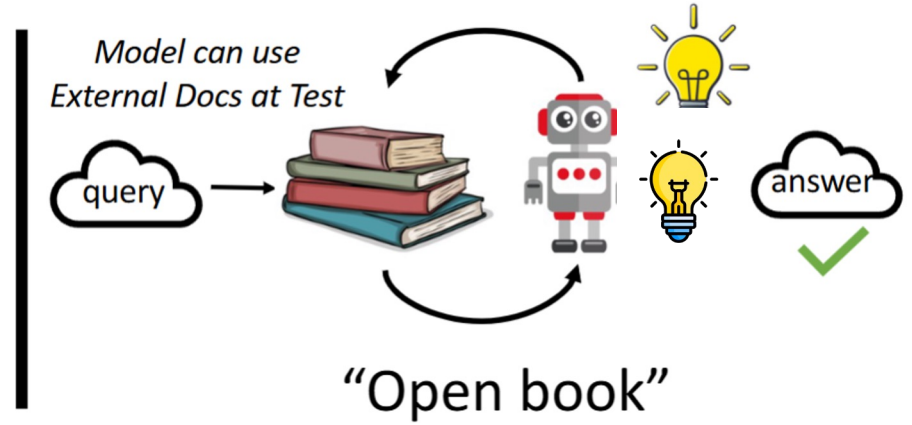
Dangers with knowledge-cutoff:

- Outdated medical information
- Security vulnerabilities
- Stale confidence (The recommended procedure is...)

Motivation for RAG



“Closed book”



“Open book”

Motivation for RAG



What LLMs don't know (the open book)

Private Databases

LLMs can't access confidential information

Hard to access information

Some information isn't widely available online, making it inaccessible to LLMs

Real time data

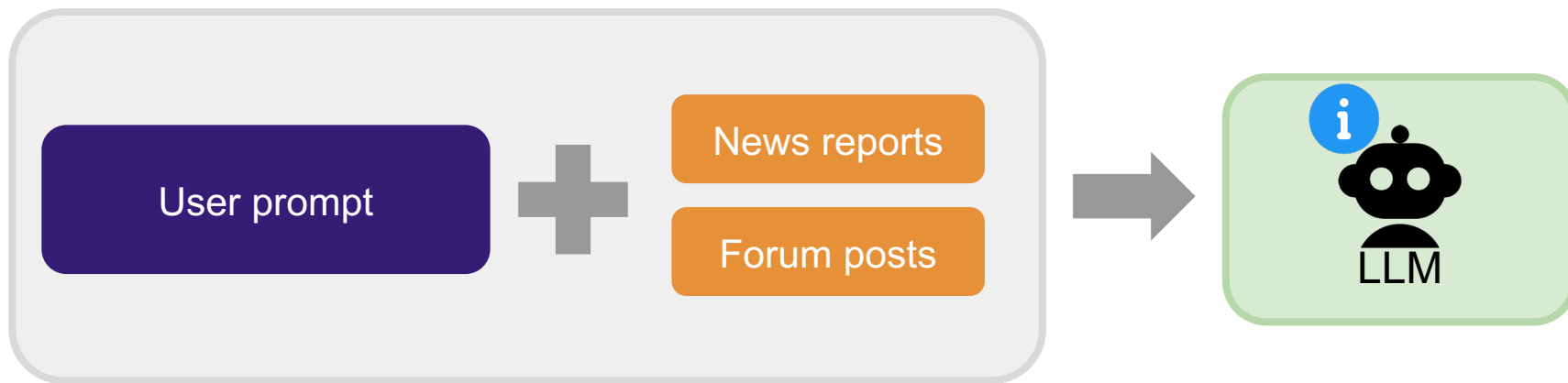
LLMs are trained on past data and don't update automatically

How to make sure the LLM knows this useful information?

Motivation for RAG



Put it in the prompt

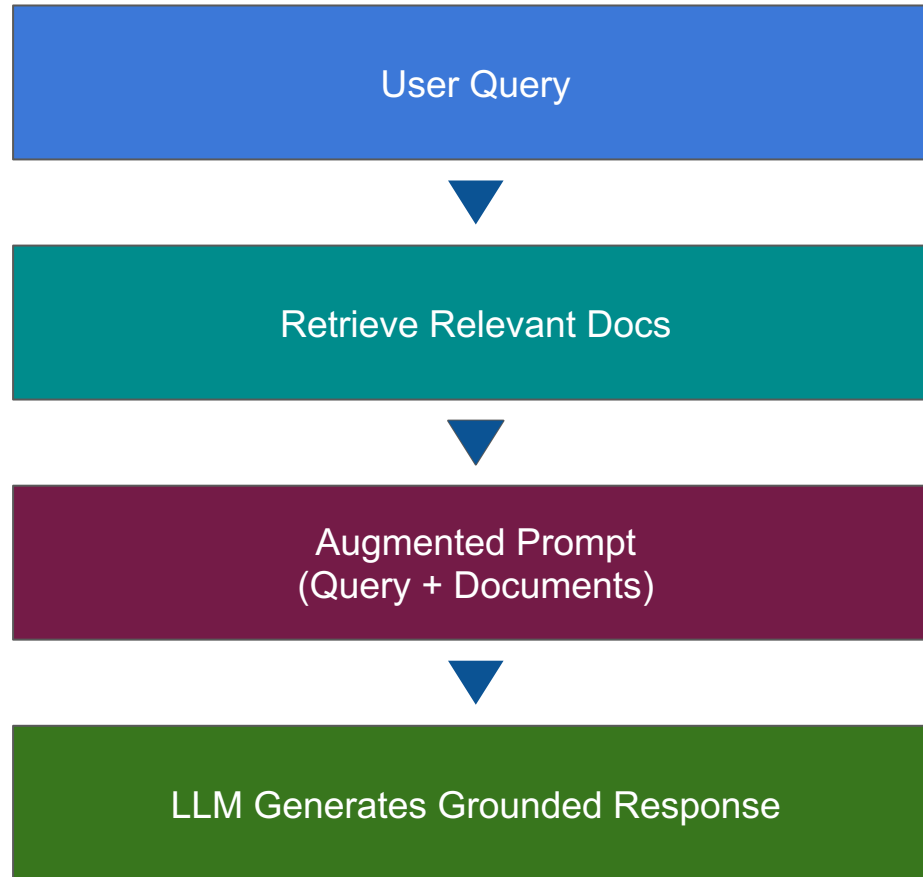




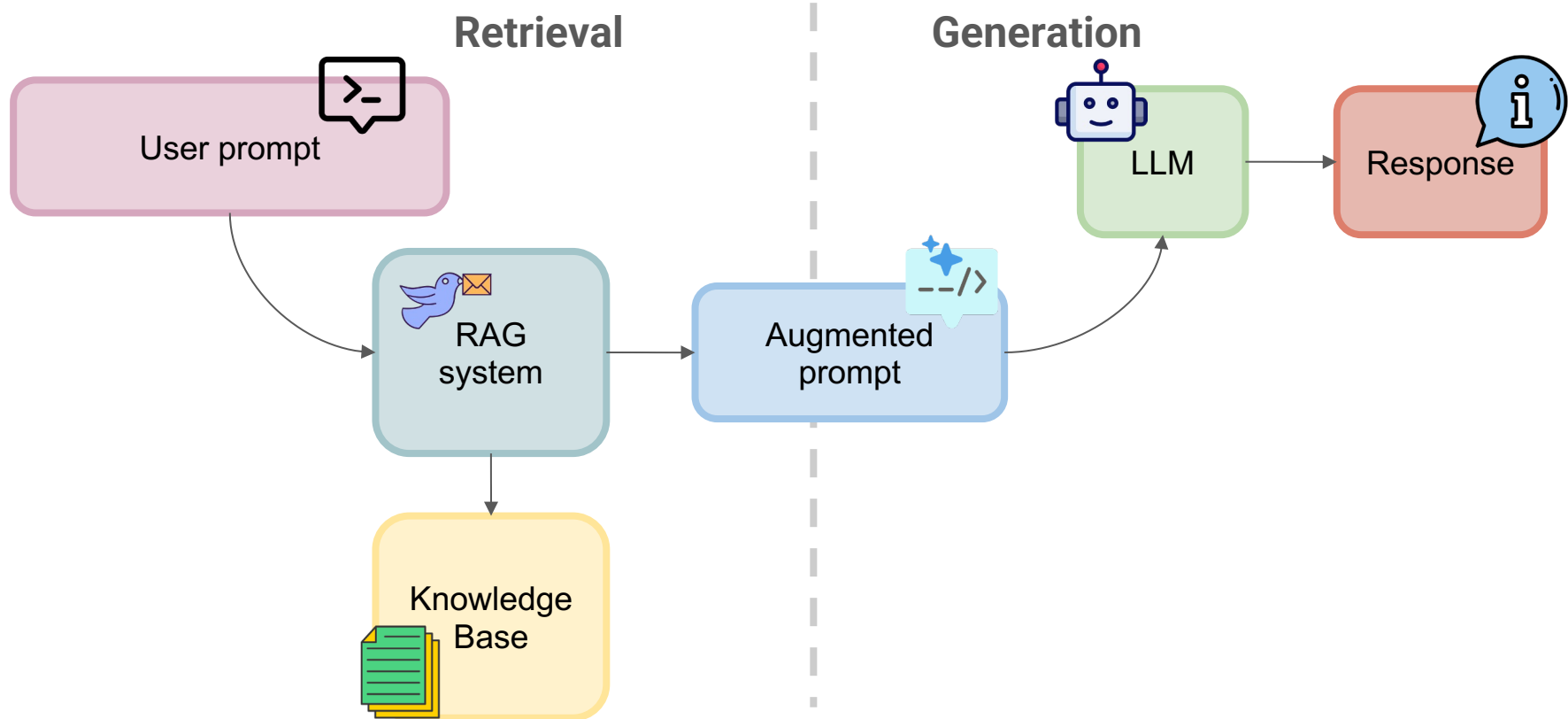
Introduction to RAG

Overview of RAG

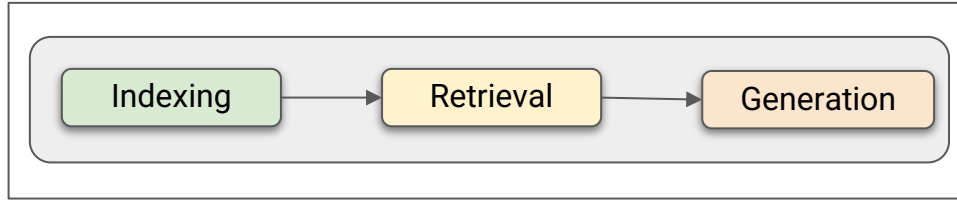
The RAG Solution



RAG architecture



RAG Workflow



Basic RAG Workflow

Motivation: LLMs can benefit from access to external knowledge sources for factual accuracy and reliability.

How (Basic) RAG Works:

- Input: User provides a query.
- Retrieval: Relevant documents are retrieved from a knowledge source (e.g., Wikipedia).
- Generation: The LLM uses the retrieved information to generate a response.

Benefits:

Factual Consistency: Improves the accuracy of generated content-

Up-to-Date Information: Allows LLMs to access current information beyond their training data.

RAG Workflow

Problem: LLMs can struggle with factual accuracy and limited pretrained information

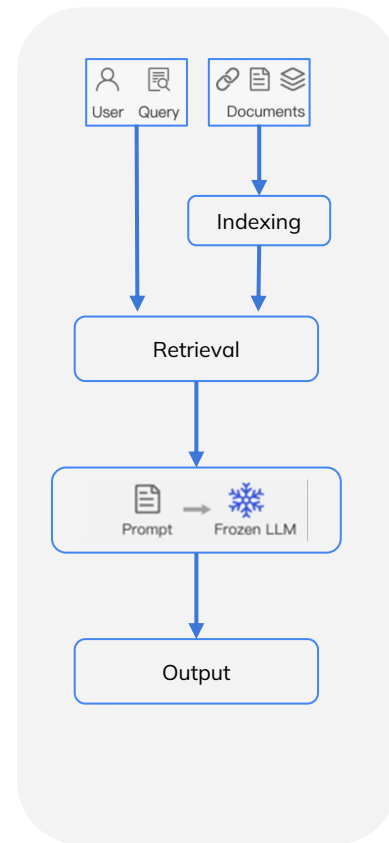
Solution: Give LLMs access to external knowledge sources.

RAG Approach:

- User asks a question.
- System finds relevant documents
- LLM uses these documents to generate an accurate and reliable answer.

Benefits:

- More factual responses.
- Access to up-to-date information.



Naive RAG

RAG Workflow

Advanced RAG: On top of the naive RAG workflow, we can add a pre-retrieval and a post retrieval step

Pre-retrieval: rewrites user query

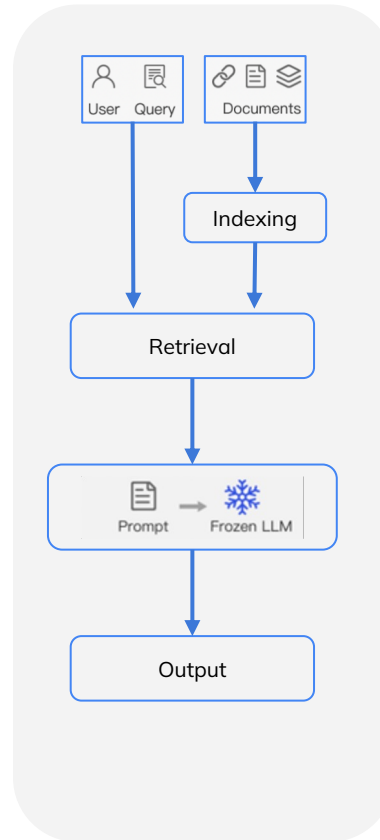
Benefits:

- Match domain-specific terminologies
- Provide additional context

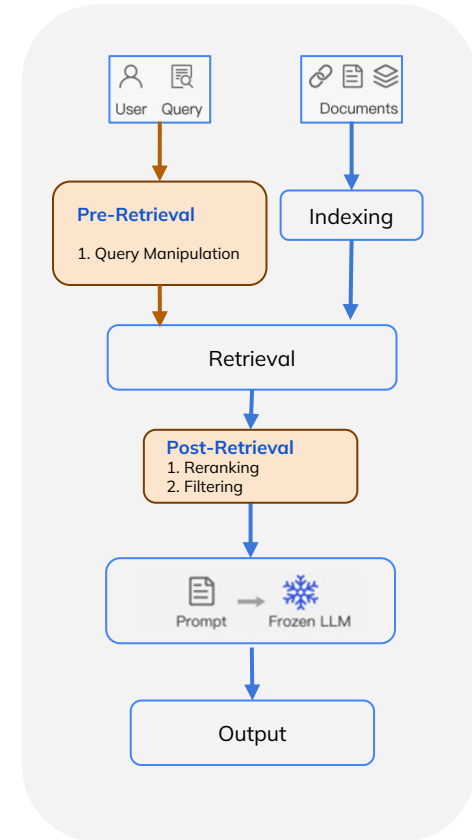
Post-retrieval: improves quality of added documents

Benefits:

- Reranks retrieved documents to reject irrelevant ones
- Filter out low-value contents



Naive RAG

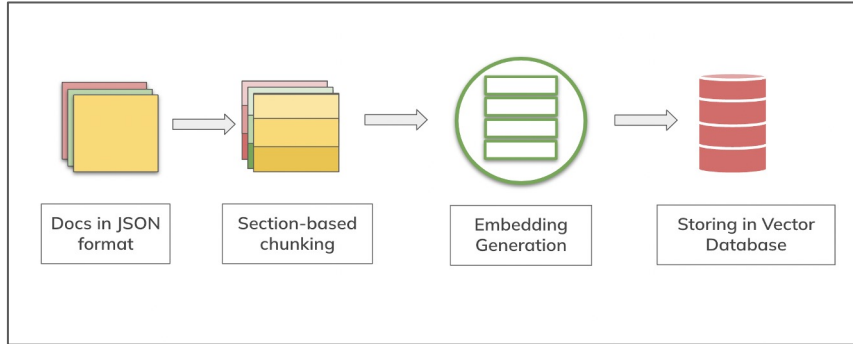


Advanced RAG

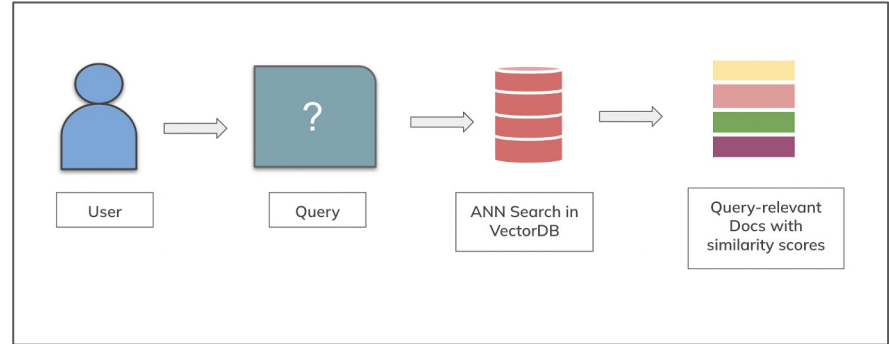
RAG: Example



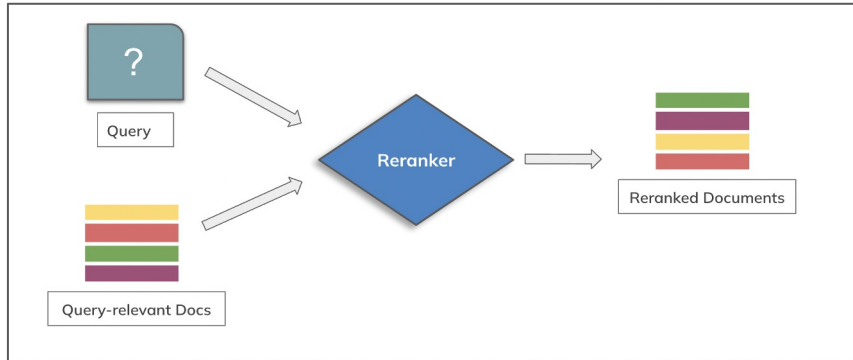
Pre-Retrieval



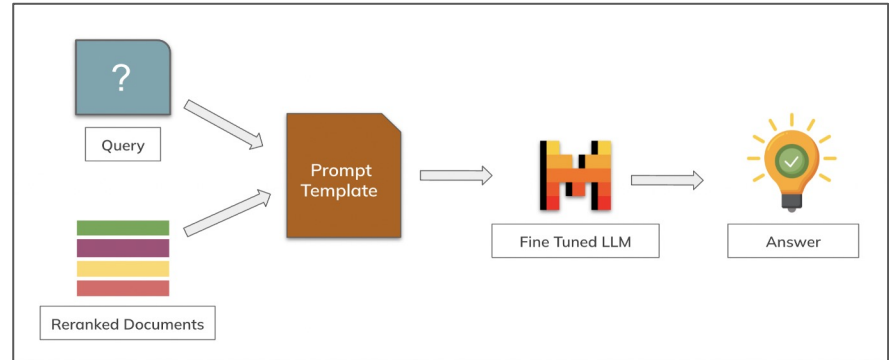
Retrieval



Post-Retrieval



Augmentation and Generation

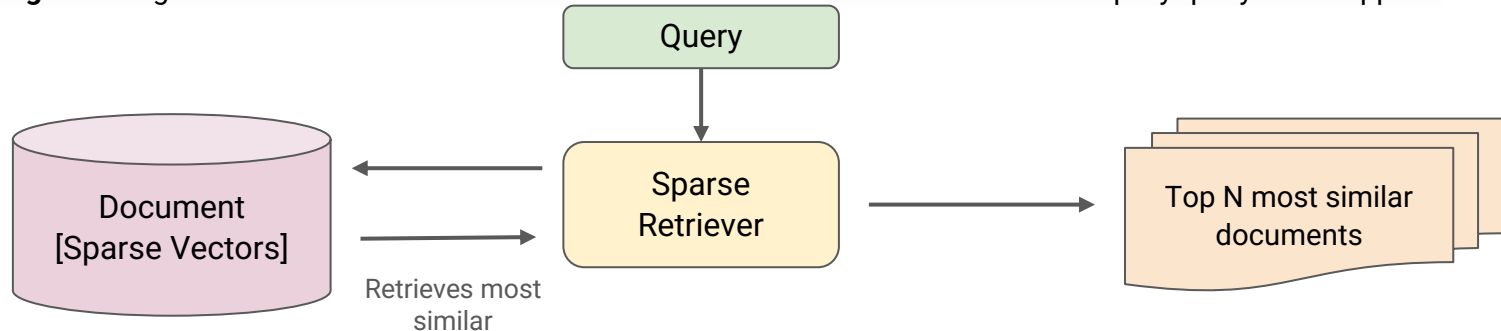


Sparse Retrieval

Sparse Retrieval in RAG: Keyword-Focused Search

Method: Matches documents based on shared keywords with the user's query.

Ranking: Uses algorithms like BM25 to score documents based on how often and how uniquely query terms appear.



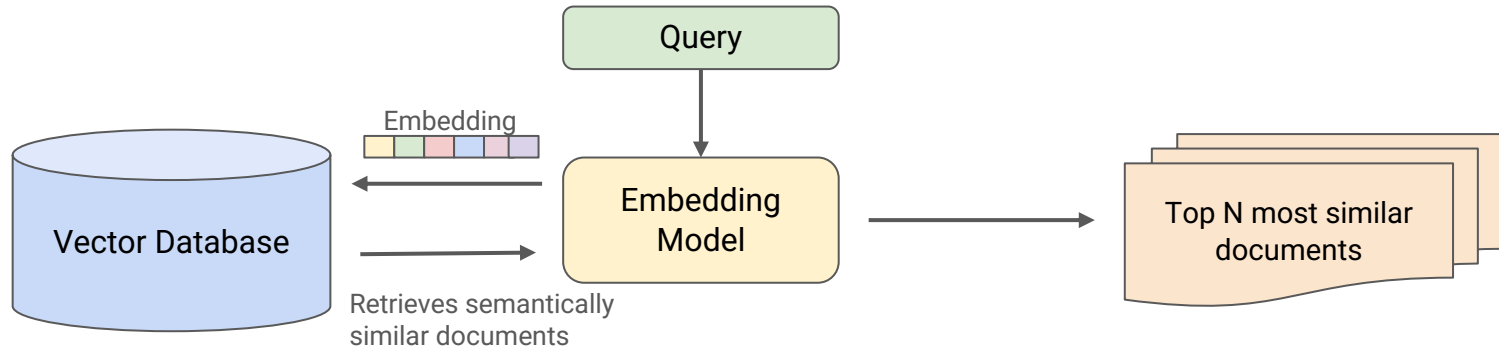
Strengths: Fast and efficient, especially for large datasets.

Weakness: Limited understanding of meaning; might miss relevant documents that use different words.

- Sparse retrieval methods are often more interpretable, as they provide clear reasoning based on term occurrences, making it easier to understand why certain documents were retrieved.

Dense Retrieval

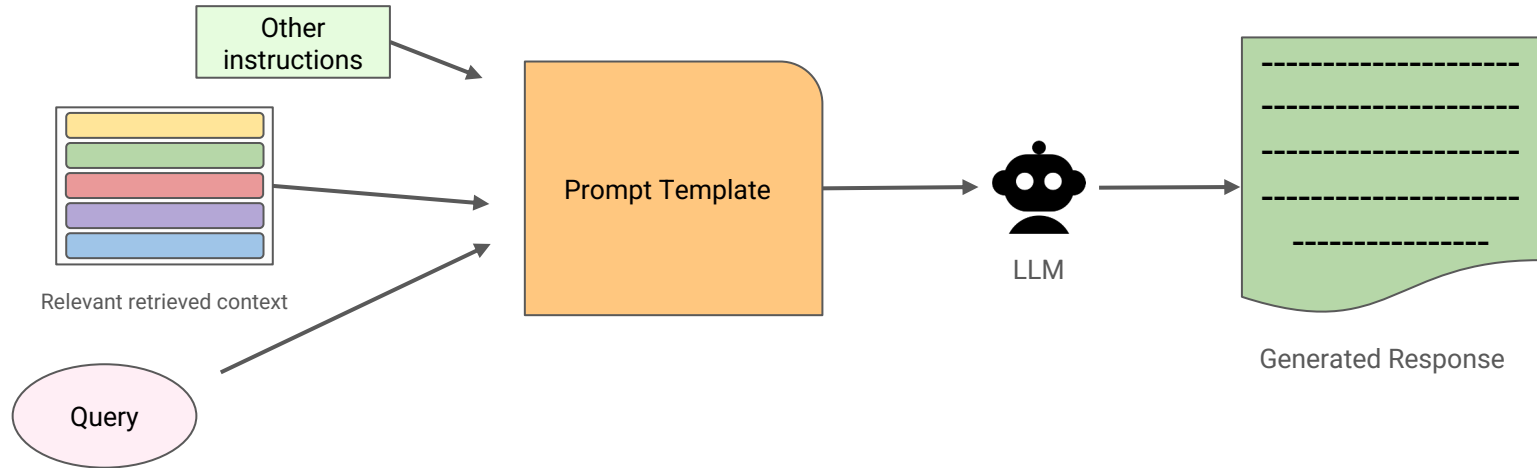
- **Method:** Uses deep learning (e.g., BERT, T5) to represent both queries and documents as vectors (embeddings) that capture semantic meaning.
- **Matching:** Finds documents with similar meanings to the query, even if they don't share exact keywords.



- **Strengths:** More accurate and relevant retrieval, especially for complex or nuanced queries.
- **Weaknesses:** Can be computationally more expensive than sparse methods.

Augmentation and Generation

- **Grounding** : Grounding involves providing LLMs with use-case specific data and knowledge sources that are not inherently part of their original training data. RAG framework acts as grounding for LLMs



- The LLM processes the input and generates a coherent and contextually appropriate response.
- This response leverages the external information retrieved in the augmentation step, allowing the model to provide more accurate and nuanced answers.



Elaborate retrieval phase

Retrieval pipeline

Raw documents are converted to retrieved chunks

1

Document Ingestion

Load PDFs, HTML, databases. Handle multiple formats.

2

Chunking

Split docs into overlapping segments (e.g. 256-512 tokens).

3

Encoding / Indexing

Embed chunks (dense) or build inverted index (sparse).

4

Query Processing

Encode the user query into same representation as the index

5

Retrieval & Ranking

Score candidate chunks (cosine sim / BM25). Return top-k most relevant.

6

Re-ranking (optional)

Cross-encoder re-ranks top-k results. Improves precision.

Chunking and Query Formulation

Chunking Strategies

Fixed-size chunking

Simple sliding window (e.g. 512 tokens, 10% overlap). Fast but may split mid-sentence.

Sentence / Paragraph

Split of natural sentence boundaries. Better coherence; relies on good sentence detection.

Semantic chunking

Group semantically related sentences together using embedding similarity shifts.

Hierarchical / parent-child

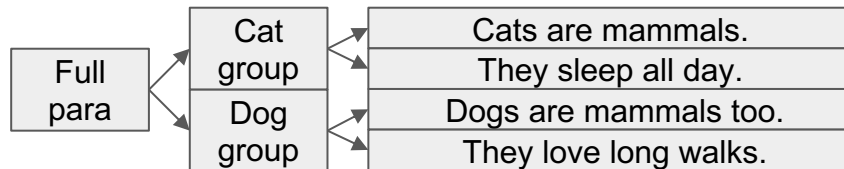
Store small chunks for retrieval, retrieve larger chunk to send to LLM (better context).

Example: “Cats are mammals. They sleep all day.
Dogs are mammals too. They love long walks.”

Cats are mammals. They sl
leep all day. Dogs are mam
mals too. They love long walks.

Cats are mammals.
They sleep all day.
Dogs are mammals too.
They love long walks.

Cats are mammals. They sleep all day.
Dogs are mammals too. They love long walks.



Chunking and Query Formulation

Query Formulation

Naive: query as-is

Send raw user question directly to retriever. Works for simple, well-formed queries.

HyDE (Hypothetical Document)

Ask LLM to generate a hypothetical answer first; use that as the retrieval query.

Multi-query

Generate multiple paraphrased queries; union the results. Handles ambiguity.

Step-back prompting

Abstract the query to a higher-level concept before retrieval. Improves coverage for complex questions.

Example: “How do I reduce my electricity bill?”

How do I reduce my electricity bill?

Lower your electricity bill by switching to LED bulbs, unplugging idle devices, using energy-efficient appliances, and adjusting your thermostat.

Ways to lower home energy costs

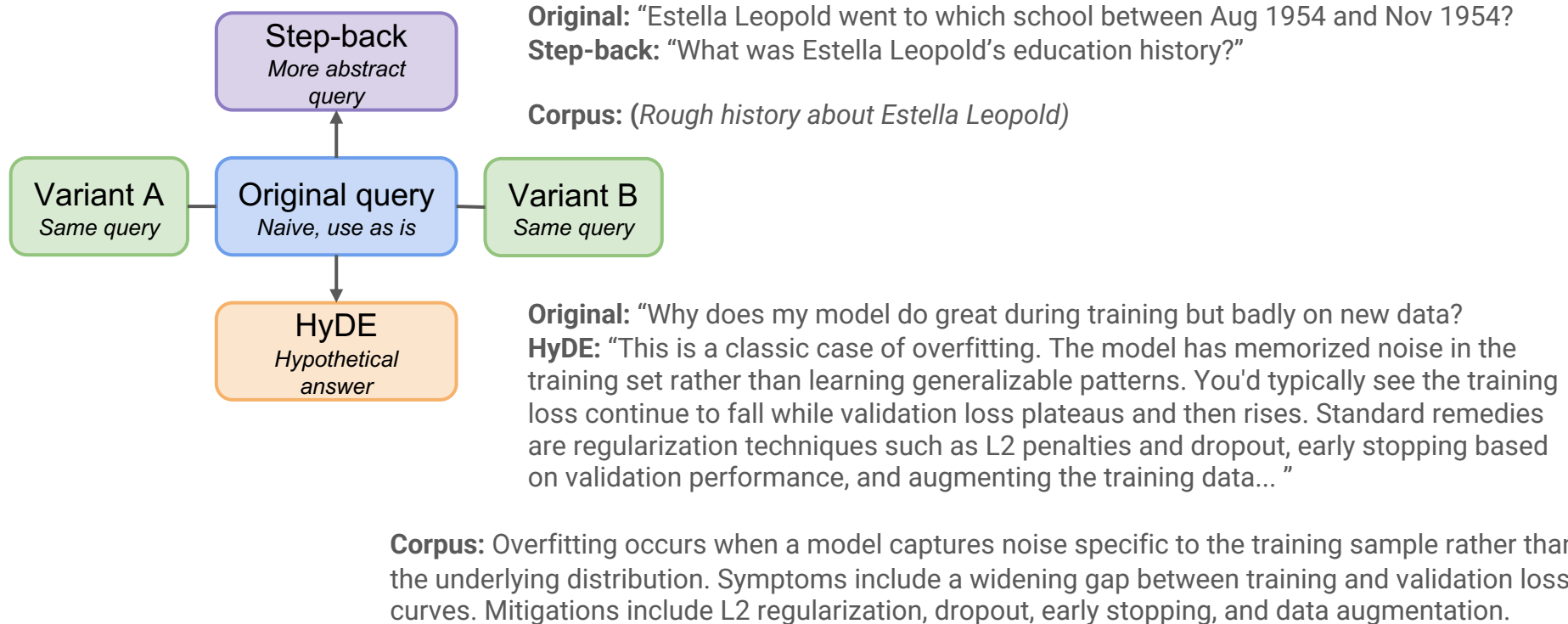
Tips to save on power consumption

How to make appliances more energy efficient

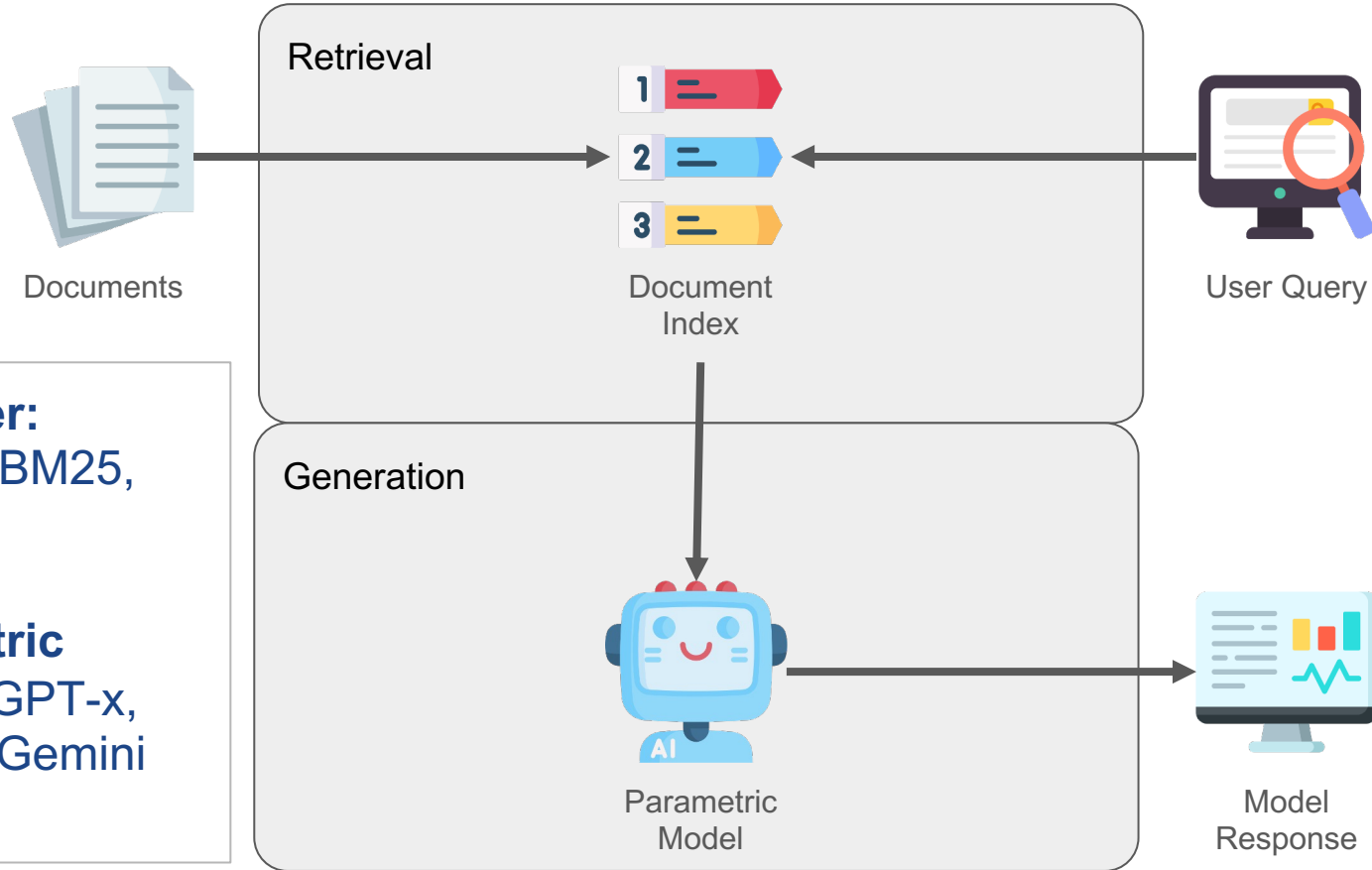
What factors affect household electricity usage?

Chunking and Query Formulation

Query Formulation: Use cases



Retriever



Retriever:
Google, BM25,
Elser

**Parametric
model:** GPT-x,
Claude, Gemini
etc.



Information retrieval methods

TF-IDF



tf-idf (term frequency-inverse document frequency)

Measure of importance of words in a document



- **Term frequency**
 - How frequent a term appears in a document
 - More frequent words are aligned with the topic of the document
- **Inverse document frequency**
 - Inversely proportional to the number of documents containing the term
 - Terms present in many documents are not significant



TF-IDF



E.g.: Consider the following three documents:



d1

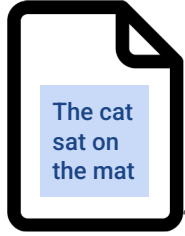


d2



d3

TF-IDF



d1



d2



d3

N = number of documents

d = current document

t = current term

df(t) = how many document contain t

$tf(t,d) = \text{count}(t \text{ in } d) / \text{total terms in } d$

$idf(t) = \log(N / df(t))$

$tf-idf(t,d) = tf(t,d) * idf(t)$

Term	TF	df	IDF	TF-IDF
the	0.333	3	0.000	0.000
cat	0.167	2	0.176	0.029
sat	0.167	2	0.176	0.029
on	0.167	2	0.176	0.029
mat	0.167	1	0.477	0.080

BM25: A Nonbinary Model

The simplest score for document d is just idf weighting of the query terms present in the document:

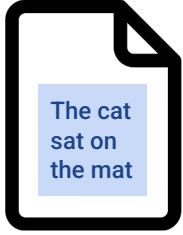
Retrieval Status Value (RSV): $RSV_d = \sum_{t \in q} \log \frac{N}{df_t}$

Improve this formula by factoring in the term frequency and document length:

$$RSV_d = \sum_{t \in q} \log \left[\frac{N}{df_t} \right] \cdot \frac{(k_1 + 1)tf_{td}}{k_1((1 - b) + b \times (L_d/L_{ave})) + tf_{td}}$$

- tf_{td} : term frequency in document d
- L_d (L_{ave}): length of document d (average document length in the whole collection)
- k_1 : tuning parameter controlling the document term frequency scaling
- b : tuning parameter controlling the scaling by document length

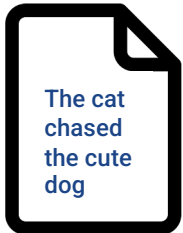
BM25: A Nonbinary Model



d1



d2



d3

cat mat

query

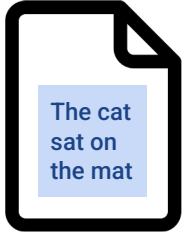
Using $k_1 = 1.5, b = 0.75, \log_{10}$.

$$k_1 \left((1 - b) + b \frac{L_d}{L_{ave}} \right) = 1.5 \left(0.25 + 0.75 \times \frac{6}{5.667} \right) = 1.5 \times 1.044 = \mathbf{1.566}$$

$$\frac{(k_1 + 1) \text{tf}_{td}}{1.566 + \text{tf}_{td}} = \frac{2.5 \text{tf}_{td}}{\text{tf}_{td} + 1.566}$$

Term	tf_{td}	2.5tf_{td}	$\text{tf}_{td} + 1.566$	factor
cat	0.166	0.416	1.732	0.240
mat	0.166	0.416	1.732	0.240

BM25: A Nonbinary Model



d1



d2



d3

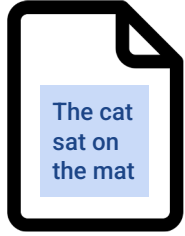
$$RSV_d = \sum_{t \in q} \log \left[\frac{N}{df_t} \right] \cdot \frac{(k_1 + 1)tf_{td}}{k_1((1 - b) + b \times (L_d/L_{ave})) + tf_{td}}$$

cat mat

query

Term	tf_{td}	$2.5 tf_{td}$	$tf_{td} + 1.566$	factor
cat	0.166	0.416	1.732	0.240
mat	0.166	0.416	1.732	0.240

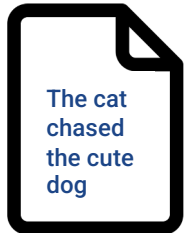
BM25: A Nonbinary Model



d1



d2



d3

$$RSV_d = \sum_{t \in q} \log \left[\frac{N}{df_t} \right] \cdot \frac{(k_1 + 1)tf_{td}}{k_1((1 - b) + b \times (L_d/L_{ave})) + tf_{td}}$$

cat mat

query

Term	df	log N/df	factor	RSV_d
cat	2	0.176	0.240	0.042
mat	1	0.477	0.240	0.114

0.156

BM25: A Nonbinary Model



d1

Using $k_1 = 1.5, b = 0.75, \log_{10}$.

$$k_1 \left((1 - b) + b \frac{L_d}{L_{ave}} \right) = 1.5 \left(0.25 + 0.75 \times \frac{6}{5.667} \right) = 1.5 \times 1.044 = \mathbf{1.566}$$

$$\frac{(k_1 + 1) \text{tf}_{td}}{1.566 + \text{tf}_{td}} = \frac{2.5 \text{tf}_{td}}{\text{tf}_{td} + 1.566}$$

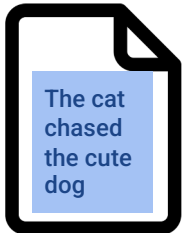


d2

cat mat

query

Term	tf_{td}	2.5tf_{td}	$\text{tf}_{td} + 1.566$	factor
cat	0.166	0.415	1.732	0.239
mat	0	0	1.566	0



d3

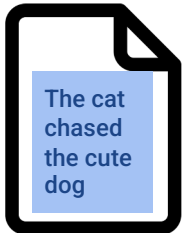
BM25: A Nonbinary Model



d1



d2



d3

$$RSV_d = \sum_{t \in q} \log \left[\frac{N}{df_t} \right] \cdot \frac{(k_1 + 1)tf_{td}}{k_1((1 - b) + b \times (L_d/L_{ave})) + tf_{td}}$$

cat mat

query

Term	tf_{td}	$2.5 tf_{td}$	$tf_{td} + 1.566$	factor
cat	0.166	0.415	1.732	0.239
mat	0	0	1.566	0

BM25: A Nonbinary Model

The cat
sat on
the mat

d1

The dog
sat on
dog

d2

The cat
chased
the cute
dog

d3

$$RSV_d = \sum_{t \in q} \log \left[\frac{N}{df_t} \right] \cdot \frac{(k_1 + 1)tf_{td}}{k_1((1 - b) + b \times (L_d/L_{ave})) + tf_{td}}$$

cat mat

query

Term	df	log N/df	factor	RSV_d
cat	2	0.176	0.239	0.042
mat	1	0.477	0	0

0.042



Common problems of RAG

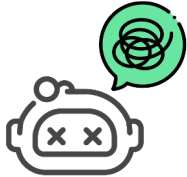
RAG failure modes



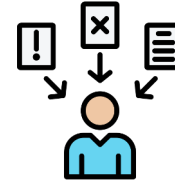
Retrieval failures



Context Failures



Generation Failures





RAG failure modes: Retrieval failures

Example Query: What is the return window for online orders?

Relevant chunk: Online orders can be returned for a refund within 30 days of delivery.

Retrieved chunk:

Wrong chunk retrieved

Semantic similarity neq relevance (vocabulary mismatch)

“Our in-store exchange policy allows swaps within 14 days”

Wrong answer: 14 days.

Incomplete retrieval

Multi-hop questions need facts from multiple docs

“Online orders can be returned for a refund”

Wrong answer: I don't know.

Stale knowledge base

Docs not updated, retriever returns outdated information

“Online return window: 15 days.” (last year's policy)

Wrong answer: 15 days.



RAG failure modes: Context Failures

Example Query: When was the company founded?

Retrieved chunk(s):

Lost in the middle

LLMs ignore chunks in the middle of long contexts

[chunk 1] Office locations...
[chunk 2] Mission statement...
[chunk 5] Founded in 2008 < correct chunk
[chunk 9] Product list...
[chunk 10] Contact info...
Answer: I don't know

Context overload

Too many retrieved chunks degrade generation quality

[5 chunks about history, HR policies, blog posts, faqs...]
Answer: vague / mixed-up

Irrelevant context

Noisy chunks mislead the LLM even when good chunks are also present

“Our partner company was founded in 1999.”
Wrong answer: 1999.



RAG failure modes: Generation Failures

Example Query: What is the refund window for online orders?

Retrieved chunk(s):

Hallucination despite retrieval

LLM still generates facts not in retrieved docs

“Refundable within 30 days”

Answer: You get 30 days, and a free shipping label plus a 10% voucher.

Knowledge conflict

Retrieved doc contradicts model’s parametric memory

Context: “30 days”

Parametric memory: “most stores use 15 days”
Answer: 15 days

Attribution errors

Model cites wrong source or ignores the provided context

“Refundable within 30 days” [Policy.pdf]

Answer: 30 days. [Source: Shipping-FAQ.pdf]

Types of RAG

- *Standard RAG*
- *RAG with memory*
- *Agentic RAG*



Standard RAG

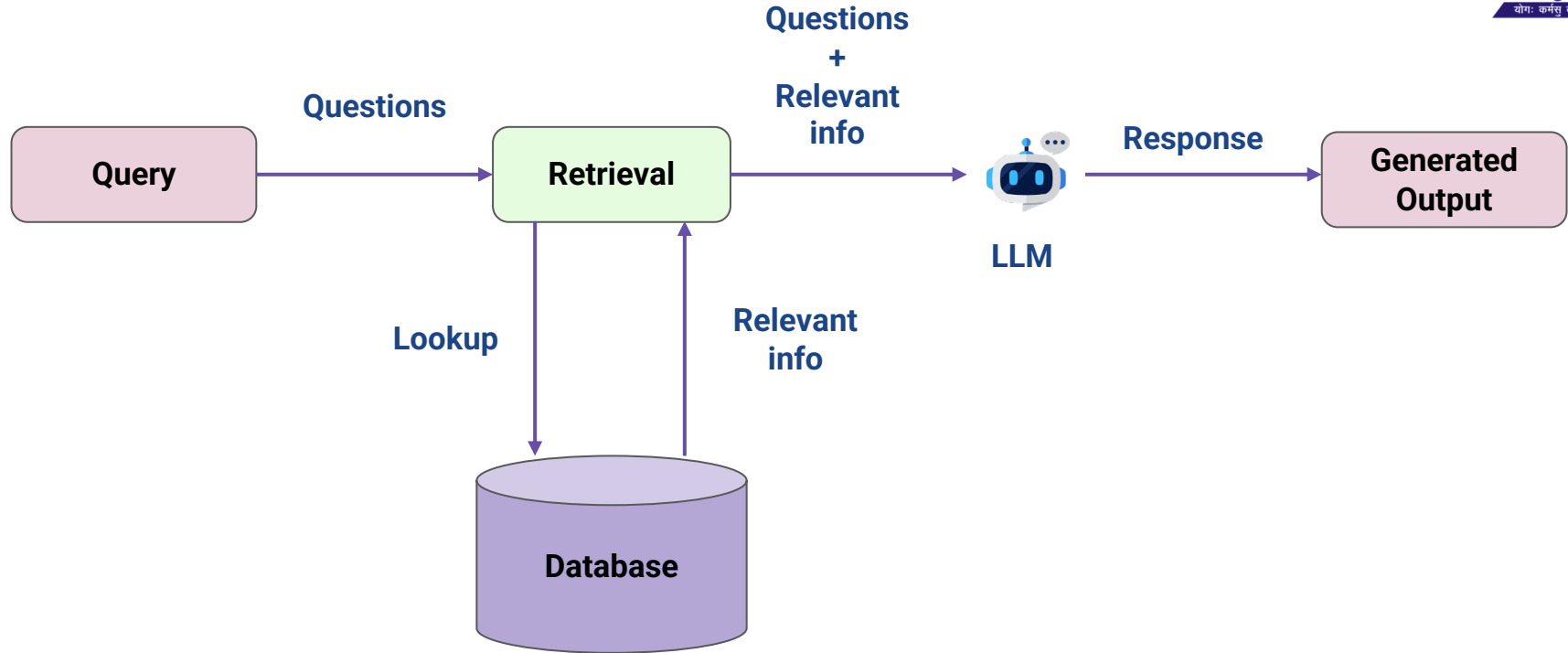
It is **Stateless**: every query is treated independently

Example:

Q: "What is the boiling point of water at sea level?"

- system embeds the query
- looks up the closest passage in the document store (e.g., a physics reference)
- LLM answers "**100°C**" grounded in that retrieved passage.

Standard RAG



Standard RAG with memory

User asks:

- Q1: "I'm a vegetarian." 🌱
- Q2: "Suggest high-protein foods."

Some use-cases:

- Personal AI agents
 - Conversational chatbots
 - Customer support systems
 - Educational tutoring platforms
-
- It adds a memory of past turns.
 - Memory is the system's ability to keep track of past interactions such as past questions, answers, or retrieved documents.
 - It doesn't just remember what was said, but understands how previous context can influence new searches.

Standard RAG with memory

Example:

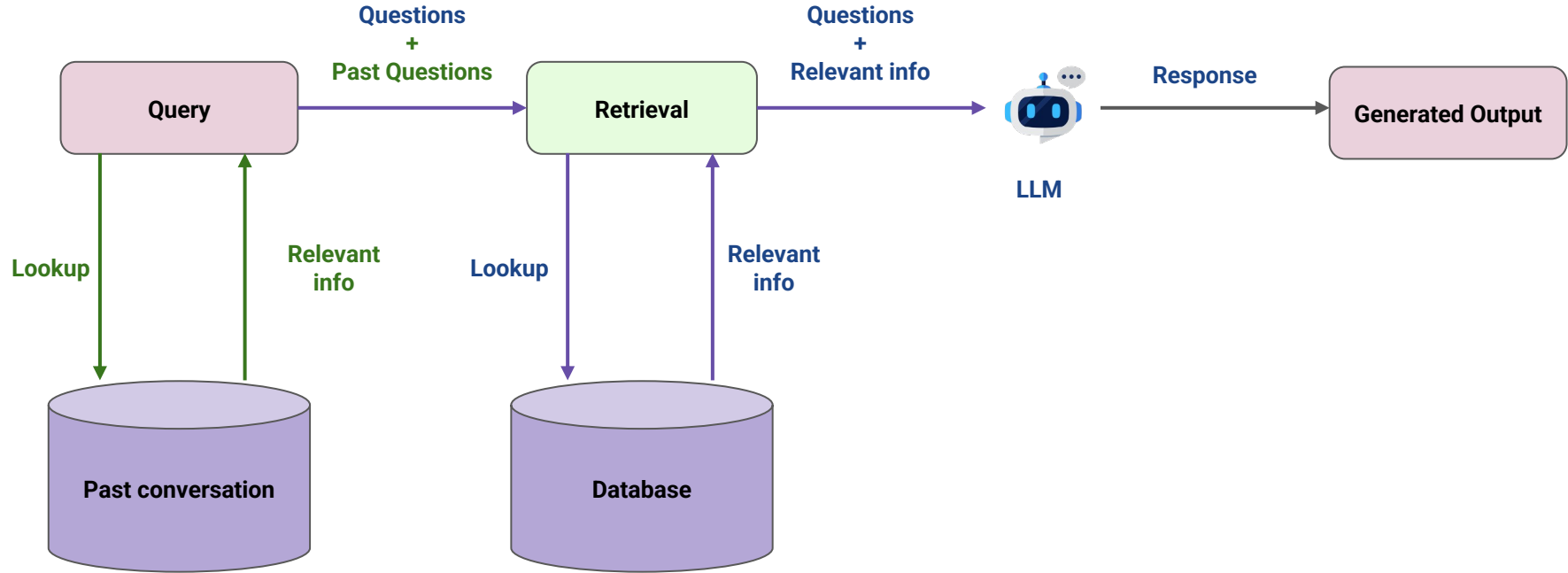
Q1: "What is the capital of France?"

A1: "Paris."

Q2: "What about its population?"

- Here, user asks about the capital of France and later refers to 'its population,'
- System recalls the context to determine the query is still referring to Paris.
- Rewrites the query to "What is the population of Paris?"
- Final answer: **Paris has about 2.05 million**

Standard RAG with memory



Agentic RAG



Example:

Q1: "How many more people live in Paris's metropolitan area than in the city proper?"

- **Steps (the agent plans and picks tools):**
 - Plan: decompose the two part, needs two figures, then do subtraction.
 - Web Search / collection lookup: city proper $\approx$ 2.1M; metro $\approx$ 11.3M. The current metro area population of Paris in 2025 is 11,347,000.
 - Reflect: "Do I have both numbers? Yes." $\rightarrow$ proceed.
 - Calculator tool: $11.3\text{M} - 2.1\text{M} = 9.2\text{M}$.

Final answer:

"About 9.2 million more people live in the Paris metropolitan area than in the city proper."

Agentic RAG

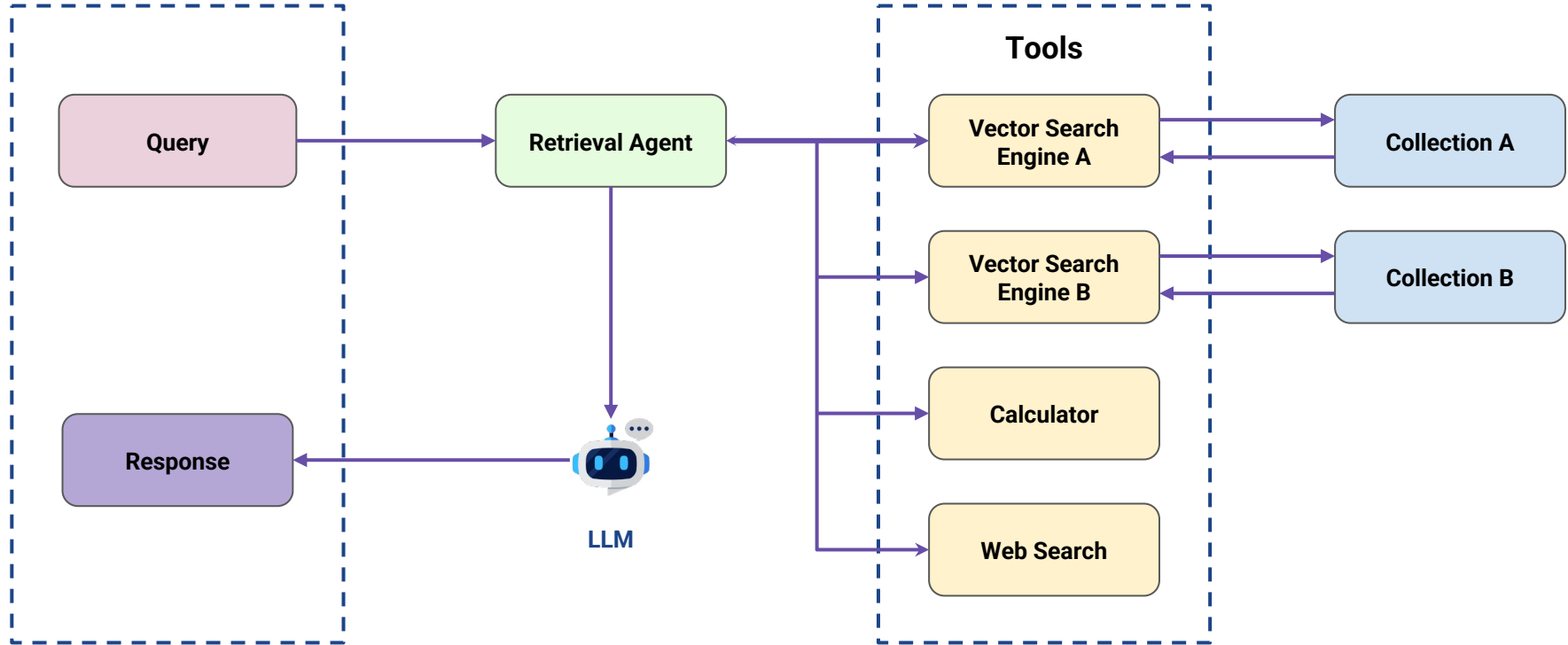


- Instead of just retrieving the first relevant documents, it plans its approach
- Decides what to investigate
- Takes action using tools.
- It breaks a task into smaller steps and searches various data sources
- Checks whether what it found answers the question, and if not, it keeps searching.

Some use-cases:

- legal research where lawyers conduct comprehensive case analysis
- financial analysis that combines market data with regulatory information

Agentic RAG





RAG: Assessment Framework

RAG boosts LLMs with external knowledge, but introduces failure points the source article identifies:

- **Retrieval issues** – wrong documents retrieved or key information missing
- **Hallucination risk** – the LLM generates content not in the retrieved documents
- **Context utilization** – the LLM fails to use retrieved information effectively

Without evaluation, these lead to inaccurate or fabricated responses which is critical in high-stakes domains like pharmaceutical research where accuracy is paramount.

Two Leading Frameworks

- **RAGAS**
- **TruLens**

RAGAS



Retrieval-Augmented Generation Assessment Suite (RAGAS)

- It is an open-source, RAG evaluation framework
- It measures how well a system retrieves and utilises external context to generate accurate, faithful and relevant responses.
- RAGAS evaluates the entire RAG pipeline i.e from context retrieval to answer generation

Answer Relevancy: Measures how closely the generated response matches the user's query intent.

Faithfulness: Evaluates whether the response stays true to the retrieved evidence.

Context Precision: Assesses how relevant the retrieved chunks are to the query.

Context Recall: Determines whether all necessary information was retrieved.

Evaluation Dataset: Represents the structured input including questions, contexts, responses and reference answers used for metric computation

RAGAS



Example:

Field	Value
question	"Who wrote Romeo and Juliet?"
contexts (what the retriever pulled)	"Romeo and Juliet is a tragedy written by William Shakespeare."
answer (what the LLM generated)	"Romeo and Juliet was written by William Shakespeare in 1597."
ground_truth (optional reference)	"William Shakespeare wrote Romeo and Juliet."

Notice: the answer adds "in 1597", which is not in the retrieved context.

That's a hallucinated detail (even if it happens to be plausible).

Watch how only one metric catches it.

RAGAS



Example:

Metric	Score	Why
Context Precision	~1.0	The one retrieved chunk is fully on-topic for the question.
Context Recall	~1.0	The context contains everything the ground truth needs (the author).
Answer Relevancy	~0.95	The answer directly addresses "who wrote it."
Faithfulness	~0.5	Two claims in the answer: "written by Shakespeare" (supported by context) and "in 1597" (not in context). Half the claims are grounded, so the score drops.

Thank You

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